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REVIEW

Review of photovoltaic cooling methods to improve electrical efficiency

Alagar Karthick[1]

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Abstract
Photovoltaic (PV) modules experience substantial electrical efficiency losses under elevated operating temperatures, driving increasing interest in active and passive cooling strategies. This review synthesizes the state of the art in PV cooling methods, categorizing them into passive techniques (heat sinks, fins, phase change materials, radiative coatings, and natural convection enhancements) and active approaches (air cooling, liquid cooling, jet impingement, sprays, and hybrid PVT systems). For each method, underlying thermal mechanisms, reported temperature reductions, and corresponding electrical performance gains are compared. The review highlights that active liquid-based cooling and hybrid PV-thermal configurations generally achieve the highest heat extraction rates but face challenges in pumping power, system complexity, and economic viability. In contrast, passive solutions offer lower maintenance and energy overhead but provide modest cooling effects and limited controllability. Emerging concepts, including nanofluids, spectral-selective surfaces, microchannel devices, and PCM-integrated designs, show promise but require further validation for durability, scalability, and outdoor deployment. Environmental and techno-economic considerations, including water use, lifecycle impacts, and cost–benefit factors, are also examined. The review concludes by identifying performance benchmarks, research gaps, and pathways for advancing deployment-ready PV cooling technologies that can reliably improve electrical efficiency under real-world conditions.

Keywords PV cooling · Electrical efficiency · Cooling methods · Photovoltaic systems · SDG 7

1 Introduction

The goal of the technology behind photovoltaic cells and modules is to lower the cost of solar electricity by improving efficiency and dependability while decreasing production costs. Solar home systems provide several benefits, including reducing environmental impact, saving money, and helping ensure a sustainable future for the Earth. Reduced energy consumption and environmental impacts from cooling systems can be achieved through solar cooling solutions, which can benefit buildings that lack sufficient roof space or orientation to support photovoltaics. Additionally, you can use them to keep structures cool during the night. Although the PCM material still saves Energy at night, its performance deteriorates over time. One easy and efficient technique for heating interior areas in cold climates and for constructing integrated PV applications is PV/T air-cooling (Paliwal et al. 2023). To lessen this impact, it is crucial to select a material that can absorb heat from the PV Module. This experiment's findings showed that solar cells may function at low temperatures while simultaneously producing water at a maximum rate (Wang et al. 2021). For the 8–13 m window region and a stable overhead air envelope, the measured cooling is in good agreement with predictions from a model that uses a locally obtained spectrum of ambient thermal emission (Harrison 1981). PV systems are viewed as adversaries when they are operational because the efficiency of PV cells, which rely on sunlight to produce electricity, decreases as the operating temperature rises (Badgujar and Sarda 2019). The absorption rate is increased, and the temperature of the solar cell is kept close to room temperature with the use of an appropriate ARC/cum emissive layer (Dwivedi et al. 2020). Their significant volume change during melting and solidification, and their poor heat conductivity, rendered any potential advantages useless. Potentially improving the low heat conductivity of open-cell metallic foams could be as

✉ Alagar Karthick
karthick.power@gmail.com

[1] Department of Electrical and Electronics Engineering, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences, SIMATS, Chennai , Tamil Nadu 602105 , India

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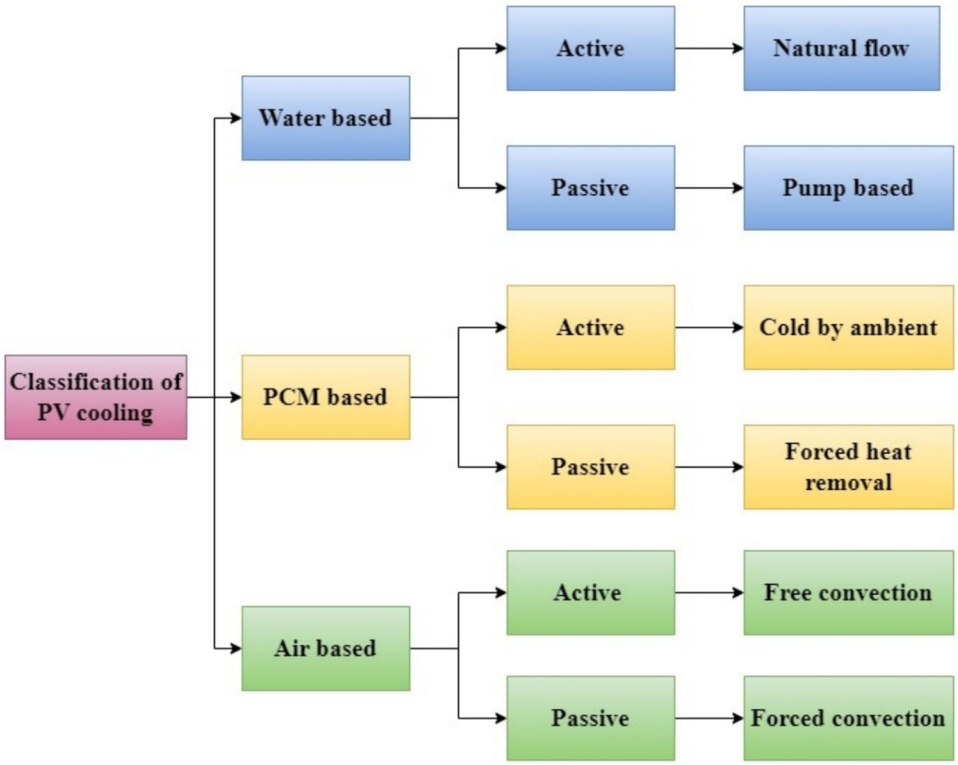


Fig. 1 Classification of PV cooling techniques

simple as adding them to the foams. To prevent PV panels from overheating, cell temperature can be reduced using cooling treatments (Dwivedi et al. 2020). To determine when the PV panels will return to their normal working temperature, a cooling model was developed (Moharram et al. 2013; Alombah et al. 2025; Venkatesh et al. 2025; Rajamony et al. 2025; Maneechot et al. 2025). To study how PV panels operate under peak solar radiation, researchers used an alternative cooling method that simultaneously cooled the panel from both sides: complete water-spray cooling (Nižetić et al. 2016). The goal is an exhaustive study of the JIC system's performance as a function of nozzle number, diameter, and the dimensionless nozzle-to-PV distance (Javidan and Moghadam 2021). To find the best pitch, thickness, number of fins, height, and tilt angle, it employed the Design of Experiments (DOE) method (Elbreki et al. 2021). Our innovative multi-layer manifold microchannel cooling system successfully reduces the cell surface temperature. It improves temperature uniformity, enabling CPV cells to produce more net output power and last longer (Yang and Zuo 2015). If you want your PV system to last and operate as efficiently as possible, you need a good cooling solution. Some effective cooling options that can reduce temperature-induced performance loss include direction, ventilation, and water- or air-based cooling systems. In doing so, it can be certain that PV systems are making a substantial impact in the renewable energy arena while running at peak efficiency. Figure 1 shows that there are numerous ways to cool PV systems, but the most common ones are water, air, and Cmp cooling. Spray cooling, natural and induced convection, heat sinks, pipes for heat transfer, and hybrid power systems are examples of the methods described in the various categories.

To make a PV system last as long as possible and work as efficiently as possible, you need a good cooling solution. Some effective cooling options that can reduce temperature-induced performance loss include direction, ventilation, and water- or air-based cooling systems. In doing so, it can be certain that PV systems are making a substantial impact in the renewable energy arena while running at peak efficiency.

2 Methodology

A comprehensive analysis of PV cooling technologies using both quantitative and qualitative evaluation criteria forms the basis of this methodology. It started by looking at all the cooling systems on the market, including active, passive, and hybrid options, as well as the performance requirements each claimed to meet. Our next step was a comprehensive analysis of critical parameters like temperature drop, electrical capacity growth, energy consumption, budgetary needs, and ecological impacts. Third, it estimated the project's viability by examining financial metrics such as operational expenses, capital costs, and payback time. Considerations such as resource utilisation, embodied Energy, carbon footprint, and life-cycle impacts were used to assess ecological

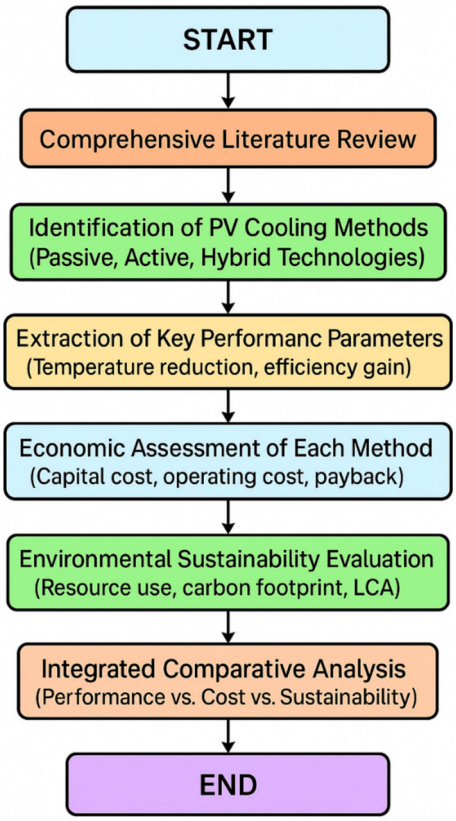


Fig. 2 Methodology adopted for the study

sustainability. It combined all the results and utilised them to construct a cohesive framework that links cooling performance to economic and environmental effects. This allowed us to develop a fair evaluation of current technologies and possible topics for future study. The study's approach is illustrated in Fig. 2.

3 PV cooling methods

By controlling panel temperature, water cooling improves the efficiency and lifespan of PV systems. Some methods for dissipating heat, increasing power output, and prolonging system life include submersion, water channels, misting, and integrated cooling tubes. Sunlight evaporates water or air through a heat pipe's evaporator and condenses it through the condenser. Three cooling solutions are utilised in the experimental endeavour to enhance the operating efficiency of PV systems. It demonstrates and examines three types of water cooling: direct backwater cooling, film water cooling, and a hybrid of the two. Figure 3 presents a comprehensive review of the relevant literature, including a comparison of solar cells with and without cooling devices to determine module temperature (Enobakhare et al. 2022).

3.1 Based on PCM

PCM cooling regulates temperature changes by using materials that absorb heat when melted and release it when solidified. This product is designed to improve energy efficiency and lower cooling costs for electronics, batteries, and buildings by effectively managing intermittent heat loads. Polycrystalline molecular condensation (PCM) cooling uses heat-absorbing and heat-releasing materials. The result is a constant temperature. PCM cooling is proper across many fields, including building cooling, electronics, and automotive applications. In photovoltaic systems, it may control module temperature, improving efficiency. Experimenting with theoretical simulations to uncover real-world applications, Fig. 4 shows a BIPV-PCM incorporated into the façade of an office building (Aelenei et al. 2014).

The use of waxes (PCMs) extends the life and efficiency of solar panels by keeping them cool. To avoid overheating, PCMs undergo phase transitions and absorb excess heat. This technology enhances panel efficiency and increases the appeal of solar electricity. To provide even better outcomes, some systems also utilise fans for active cooling (Alqatamin and Jinzhan 2025). Figure 5 shows that one PV module was designed to be cooled by the active cooling system, while the other module served as a control. This study aims to investigate and explain the effects of PCM and NPCM on PV cell cooling, as PCM is the most efficient material for capturing heat from solar PV systems (Alqatamin and Su 2025a). One litre per minute (L/min) was determined to be the most realistic heat-extraction flow rate in a performance comparison (Jakhrani et al. 2017).

Half of the Energy from sunlight is converted into heat and half into electricity, which lowers the efficiency of photovoltaic cells. Various cooling technologies are compared in Fig. 6. To do an accurate comparison, you need to know the global cooling value, the capacity you obtained, the relative and overall efficiency increases, and the method you used to cool (Mohammed et al. 2023).

Using PV/T modules, the thermoelectric device can cool in the summer like a Peltier device. With the chilly side facing inside, heat can be absorbed, resulting in cooler interiors (Jakhrani et al. 2017). Renewable Energy is more important than ever before due to the growing population, the diminishing supply of fossil fuels, and the undeniable impact of climate change. Shows how the average power generation in the experiment increased by 7% (Alqatamin and Su 2025b). Proper cooling makes solar panels last longer and perform better, which in turn extends battery life and keeps energy output high. From basic fans to liquid circulation, there are a variety of cooling methods that can be used to achieve the results shown in Fig. 7 (Kasaeian et al. 2017).

Depending on the material used, solar cells convert sunlight into electricity with efficiencies ranging from 7 to 40%.

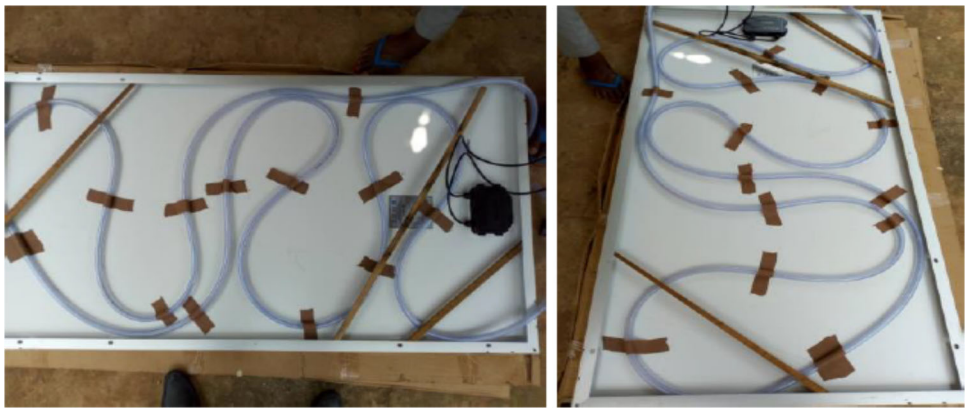


Fig. 3 Examining the PV modules' water cooling system (Enobakhare et al. 2022)



Fig. 4 System for cooling the façade using photovoltaic panels (Aelenei et al. 2014)



Fig. 5 System for active cooling that was manufactured (Qayoom Jakhrani et al. 2017)



Fig. 6 The Fresnel reflector with PV/T collector (Amanlou etal. 2018)



Fig. 7 A PV/T hybrid collector ((Kasaeian et al. 2017)

You can protect them from overheating and ensure they continue to perform efficiently by spraying water over the front surface (Yogesh et al. 2016). An independent thermal unit may directly absorb sunlight, unlike a solar panel. While testing has shown that panels submerged in water can capture up to one-third of the sun's rays, this comes at the cost of reduced electrical output (Alqatamin and Su 2025). Solar panels lose efficiency and lifespan when exposed to high temperatures. Concentrator photovoltaic (PV) systems, which are gaining popularity and employ multi-junction cells for increased efficiency, require specialised cooling solutions to maintain Fig. 8 (Xiao et al. 2018)—microencapsulated PCM layers in PV panels combined Fig. 9 (Al-Waeli et al. 2017).

Thermal management strategies for solar panels are currently under evaluation.

Due to heat loss, solar panels become less efficient. Lowering power output is possible. Methods that keep panels cool, such as tracking systems and mirrors, can increase efficiency

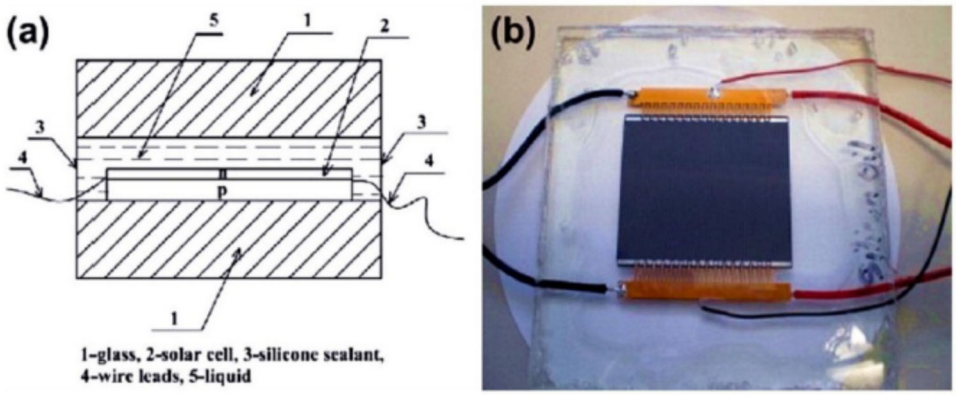


Fig. 8 **a** Micro-module form factor, **b** Silicon solar panel including a single micro-module (Xiao et al. 2018)

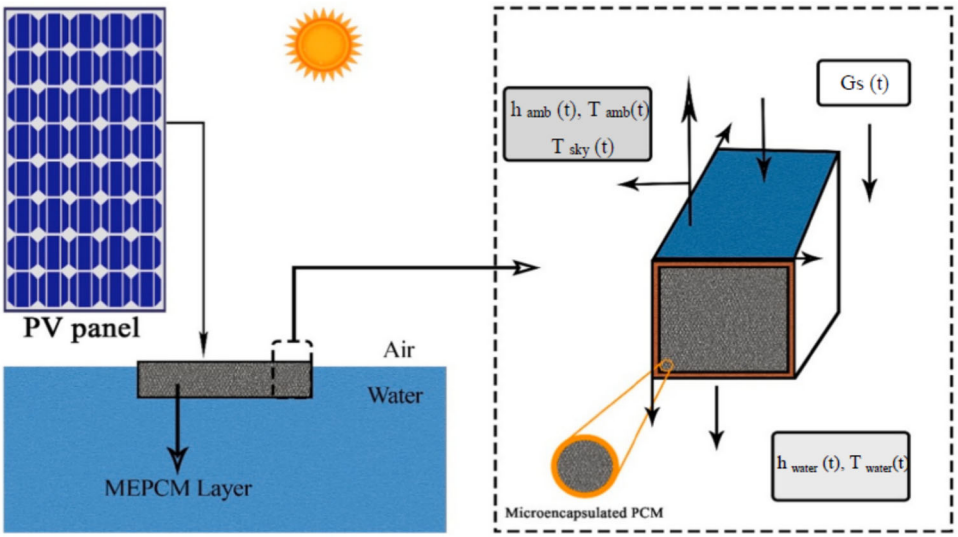


Fig. 9 Microencapsulated PCM layers in photovoltaic panels (Al-Waeli et al. 2017)

(Alsaqoor et al. 2023). The use of photovoltaic (PV) panels might alleviate South Africa's refrigeration problems and India's power outages. Insulation provides additional cooling and sunlight-efficiency benefits (Alsaqoor et al. 2023). The term "PCM-based BIPV/T system" refers to a combination of solar panels and heat storage materials. These systems do triple duty, producing power, storing heat, and controlling the temperature inside the structure. Saving money, using renewable Energy, and creating a more pleasant building environment are all outcomes of this.

3.2 Air-based PV cooling

Using either natural or artificial convection, airflow cooling removes heat from components. Through the circulation of cooled air, it maintains appropriate temperatures, ensuring the reliability of components. Used extensively due to its adaptability, affordability, and ability to withstand moderate heat loads. The process of heat transfer by natural convection is facilitated by the motion of fluids propelled by buoyant forces. It is not dependent on mechanical devices such as pumps or fans, unlike forced convection. This happens in both artificial and natural systems, such as electronic gadgets, ocean currents, and air circulation. To optimise thermal management and build effective heat transfer systems in diverse engineering applications, it is vital to understand natural convection (Alqatamin et al. 2025). Figure 10 depicts air base BIPV/T systems, which are solar panels that seamlessly integrate with structures. It is aesthetically pleasing, Energy efficient, and produces heat and power. These systems have the potential to make airbases Energy independent while also being beneficial to the Environment. Adding polycrystalline materials (PCMs) to solar panels increases their heat storage capacity and reduces their temperature rise (Čurpek and Čekon 2020).

Instead of relying on natural processes, active cooling uses refrigerators, pumps, or fans to remove heat from a system. To avoid overheating and guarantee peak performance, this forced circulation is crucial for powerful electronics. As a proactive measure, solar panels employ thermoelectric coolers to control heat. These coolers reduce cell temperature and increase efficiency by converting heat into power.



Fig. 10 Outdoor BIPV façade test cell on a large scale (Čurpek and Čekon 2020)

Compared with more conventional systems, thermoelectric coolers excel due to their use of direct current (DC) power, portability, reliability, lack of moving parts or leaks, and ability to switch between heating and cooling. Changing the direction of the current allows you to utilise it for both cooling and heating (Alqatamin and Su 2025). Shade, ventilation, and thermal mass are among the design tactics and materials used in passive cooling, which relies on natural processes to maintain a comfortable temperature without electricity. In milder areas, this green method can be helpful because it reduces air conditioning use. Cooling that doesn't use moving parts, such as fans or pumps, is called passive cooling or natural-flow cooling. Components are cooled using a circulation system that alternates between warm and cold air. When it comes to sustainable design, technology, and structures, this low-maintenance method is perfect. To actively circulate liquid coolant to absorb and remove heat from critical components such as computers, machines, and cars, pump-based cooling is used. This regulated flow keeps components cool and improves their performance and longevity, while also ensuring effective heat transfer. About 10–15% of the system's overall energy use came from the pumps, which PV and battery systems could power. To increase efficiency and possibly reduce energy costs without sacrificing thermal comfort for occupants, an optimal control method was implemented (Odufuwa et al. 2024). The system's cooling mode reduces the temperature of the PV cells and the demand on the building's air conditioning system by pre-cooling the air that enters the curved PV ventilated façade using the heat pump evaporator. According to Fig. 11, performance and flexibility could be significantly enhanced by combining energy storage systems (Cai et al. 2024).

Improved solar energy efficiency and less soil temperature imbalance are two benefits of the PV/T system. The temperature-difference control method outperformed the time-control method in long-term operation (Li et al. 2024). When looking at fins that are the same in volume and cross-sectional area, the pin–fin arrangement is clearly superior.

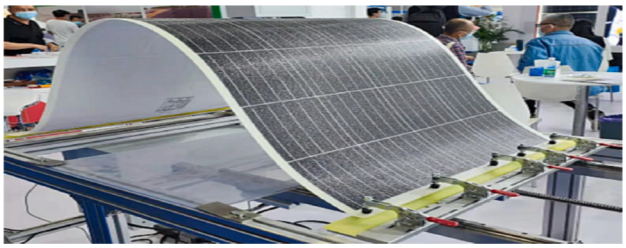


Fig. 11 Module for flexible thin film photovoltaics (SUNMAN) (Cai et al. 2024)

By the 20-min mark, the PV-PCM case devoid of pin fins has increased by 46%. Places with naturally cold climates are ideal for ambient cooling, as they remove heat from the air by utilising the cool air around. With this passive method, data centres, buildings, and even some industrial operations can reduce their energy use and expenses. Industrial, automotive, and electronic equipment can actively dissipate heat using forced-air or liquid cooling, which uses fans, pumps, or blowers. Optimal performance in harsh conditions is impossible without this vigorous circulation, which keeps things cool and prevents overheating.

3.3 Spray cooling

To manage the temperature and dissipate heat, PV modules can be "sprayed" with coolant. The working temperature of the solar cells is reduced as the evaporating coolant removes heat from them. This method increases energy conversion efficiency and prolongs module lifespan. Improving the long-term viability and efficiency of spray cooling requires additional research. According to experimental results, the electrical efficiency of traditional solar panels decreases as temperatures rise (Kasaeian et al. 2017) The impact of cell temperature on photoelectric conversion efficiency under various spray conditions was accounted for in a 3D model

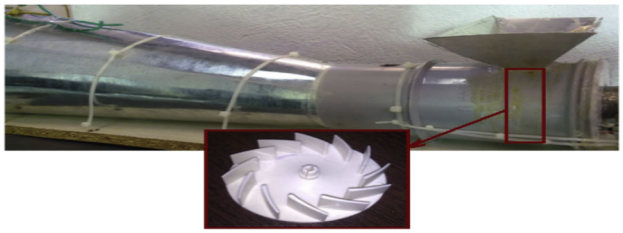


Fig. 12 The modified cooling system and the form of the turbine blade (Yang and Zuo 2015)

that simulates mass and heat transmission. For optimal cooling, use the nozzle at the bottom of the tank (El Alami et al. 2025a, 2025b; Selvaraj et al. 2025; Karthick et al. 2025).

4 Emerging cooling methods

Emerging cooling technologies are using innovative approaches to combat the growing heat problem. Improved heat spread by graphene, effective heat removal by phase-change cooling, and superior heat transfer by microchannel heat sinks are all examples of such materials. Nanofluids are improving heat transfer in a wide range of applications, including as coolants. Improving efficiency, performance, and dependability in demanding applications is the goal of combining these technologies in hybrid systems, another fascinating development.

4.1 Jet impingement PV cooling

By penetrating the thermal boundary layer with high-velocity fluid jets, jet impingement cooling dissipates heat quickly and uniformly. Electronics, turbines, and heat exchangers can all benefit from this method because of its superior performance in dealing with large heat loads, improving cooling efficiency, and keeping surface temperatures low. As shown in Fig. 12, the conic tunnel wind collector can cool photovoltaic cells and generate power using a turbine. Because it converts wind energy straight into electricity, this novel cooling technology is ideal for photovoltaic solar cells. It is also taken into account how the turbine blade affects the cooling effectiveness within the conic chamber (Yang and Zuo 2015).

In a flat-plate concentration solar system, the goal of the mini-channel/jet impingement heatsink model is to reduce the surface temperature of a triple-junction solar cell. The efficiency of solar cells can be improved by reducing the average temperature and increasing thermal homogeneity, which is achieved by raising the pin fin angle from zero to forty degrees (Enobakhare et al. 2022).

4.2 Cooling using geothermal PV electricity

Geothermal cooling is a method that uses the Earth's natural cooling. Subterranean fluid pipes remove heat from the Environment and send it to a colder location. By reducing reliance on conventional cooling, this energy-efficient, eco-friendly technology is cost-effective and ideal for a range of applications. Using an EnergyPlus model, this analysis determines the building's heating and cooling loads and the effects of several strategies for controlling the temperature setpoint. Optimising the building's cooling and heating times and energy output from its PV roof will be the subject of future research on new solutions for internal temperature control (Priarone et al. 2024). Renewable energy variability is a common problem, and there is a lack of research on how to integrate significant shares of renewable Energy in the region. The cooling system's efficiency was evaluated at an isolated PV facility with single-axis solar tracking (Oyewo et al. 2024). On the other hand, the combination of geothermal and PV power could accelerate and reduce the costs of Costa Rica, Honduras, and Guatemala's energy transition. Figure 13 shows the results of the year-long test campaign conducted across different environments, demonstrating that the procedure is technically feasible and may increase the efficiency of solar modules (Lopez-Pascual et al. 2024).

To improve the efficiency of PV panel rear-surface cooling, this alternative cooling system uses an earth-to-air heat exchanger (EAHE) to first chill the surrounding air. As shown in Fig. 14, the EAHE system can reduce the surrounding air temperature by up to 14 °C, whereas the soil temperature ranges from 3 to 25 °C (Elminshawy et al. 2019).

4.3 Fin cooling

Adding fins made of copper or aluminium on the back of PV modules improves their convective heat dissipation capabilities. Figure 15 demonstrates that fins extend the life of a system and increase power output by keeping solar panels cool. To make the most of this advantage, new fin designs and applications are essential (Elbreki et al. 2021).

4.4 Submersion cooling using liquids

Bypassing air cooling, liquid immersion cooling dissipates heat more effectively by submerging electronics in a heat-conductive liquid. Particularly useful for high-power data centres, this approach optimises performance by maintaining optimal temperatures, resulting in reduced costs, energy savings, and a smaller system footprint. In addition to developing a control chart for managing SPILC system parameters, it found a correlation between the coolant's thermal characteristics and the system's performance. The performance of the SPILC system is maximised under constant conditions when

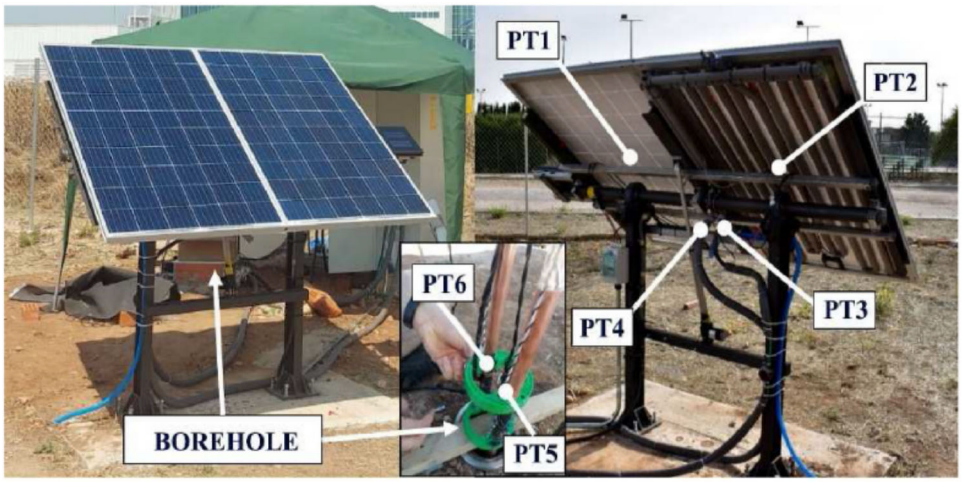


Fig. 13 The experimental setup's primary temperature sensors (Lopez-Pascual et al. 2024)



Fig. 14 Snapshot of the experimental apparatus (Elminshawy et al. 2019)

the immersion liquid cabinet is operated with anti-gravity flow direction (Huang et al. 2024). Optimising the structural features of an immersion-liquid-cooling cabinet improved temperature uniformity and eliminated hot spots, according to this study. The model was fine-tuned with respect to the distribution of temperatures by utilising fixed internal structural parameters that centre on the effects of fluid flow routes (Shehram et al. 2025a). A more consistent temperature across the battery pack was achieved by raising the coolant flow rate. To maintain temperature differences below 5 °C, it is recommended to use a flow rate of at least 0.5 L/min. Using both computational and experimental methods, researchers examined how different flow rates affected the immersion-cooled battery pack's heat dissipation and temperature uniformity (Wu et al. 2024).

4.5 Cooling via evaporation

The results show that, compared to an uncooled PV-STEG system, the PEC system's operating temperature is much lower. Thermoelectric effects, heat transfer, and moisture transport were all integrated into a COMSOL-based multiphysics coupling model (Gao et al. 2023). Compared with a conventional DPEC-based method, the proposed system demonstrated superior cooling performance and increased PV energy efficiency. Because of the potential inefficiency of evaporative cooling in humid environments, pre-dehumidification of the air is often necessary(Yang et al. 2024). Space is saved compared to separate cooling systems by integrating a PV panel with an evaporative cooling system beneath it in the unique PV/EC system. No leaking was experienced thanks to a 0.3 L/min adjustable water flow rate and inlet temperatures ranging from 25 to 35 °C. A linear relationship, similar to that observed in solar collector testing, is

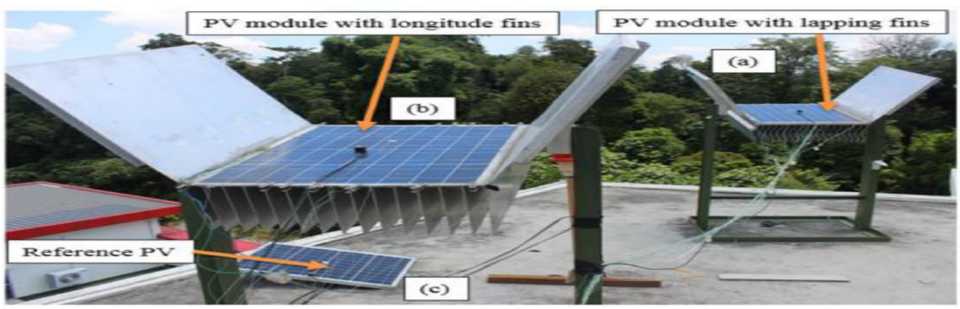


Fig. 15 **a** A PV module including a planar reflector with lapped fins, **b** A PV module with planar reflectors attached to its longitudinal fins, **c** A solar panel devoid of cooling fins (Elbreki et al. 2021)

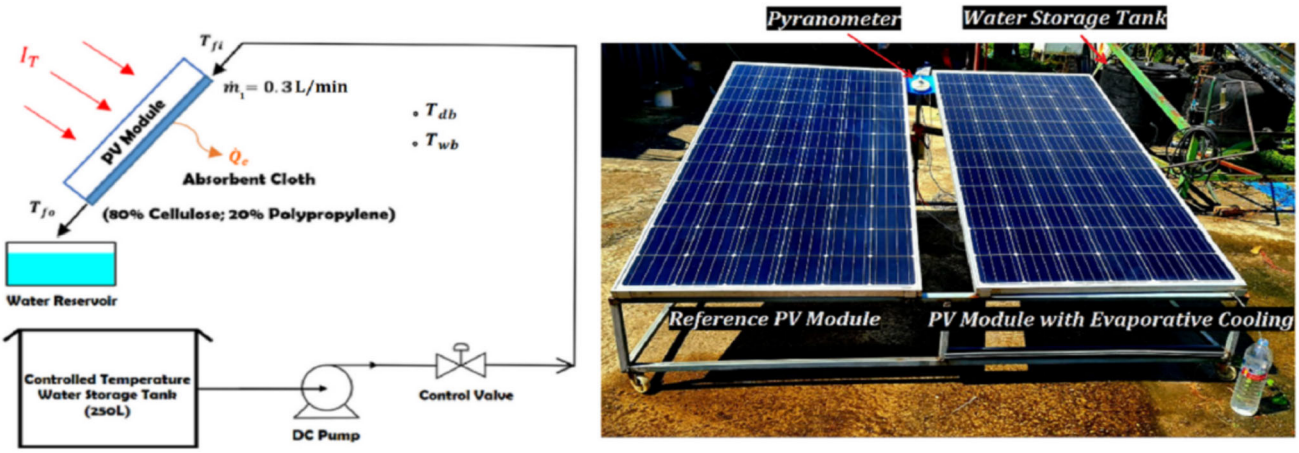


Fig. 16 Images of the tested PV modules, both with and without evaporative cooling, and a schematic of the PV module using evaporative cooling (Chea et al. 2023)

shown in Fig. 16 for the thermal performance of the PV module with evaporative cooling, evaluated by total thermal efficiency and temperature changes relative to incident solar intensity (Chea et al. 2023).

4.6 Cooling of heat pipes

Heat pipe cooling is an efficient method of transferring heat that uses the phase change and capillary action of liquids. As the liquid passes from the hot end to the cold end, it evaporates, condenses, and then returns to the hot end, allowing for rapid heat dissipation. Excellent for LED systems and high-performance electronics. Because heat pipes remove wasted heat, using them to cool PV panels is inefficient. Heat-pipe integration with water-heating systems increases electrical output and thermal gain. The coating of the vacuum tube absorbs solar radiation, which is then transformed into thermal Energy and delivered to the heat pipe through an aluminium fin. A low-material PV cooling method that is both cost-effective and efficient is air cooling (Wu et al. 2024). Future improvements in PV system cooling methods and suitable design components for big solar plants are depicted in Fig. 17 (Alobaid et al. 2017).

4.7 Thermoelectric microchannels

To prevent solar cells from overheating, microchannel PV cooling circulates water or a water–glycol mixture through small channels near the cells. Especially in extreme heat or direct sunlight, it increases power output, prolongs module lifespan, and makes them more reliable (Yang and Zuo 2015). Microchannels with one header and those with multiple headers both have hydraulic diameters of 0.667 mm. The cooling mechanism was operated by means of water. Continuous research aims to improve microchannel cooling systems for PV applications by reducing manufacturing costs, increasing system longevity and reliability, and boosting heat transfer efficiency. By maximising the efficiency and lifespan of PV modules, microchannel PV cooling can contribute to solar Energy's continued success as a clean and sustainable power source (El Alami et al. 2025b, 2025c, 2024; Alami et al. 2025a, 2025b, 2025c). Heat sinks made of high-thermal-conductivity materials allow solar panels to operate more efficiently. To keep the panels cool and avoid performance decreases, these fins function similarly to expanded surfaces, dissipating heat. The effectiveness of a cylindrical pin fin heat sink for cooling solar panels using natural air circulation was demonstrated in this study. The heat sink was angled downward and constructed of coated aluminium (Alsaqoor et al. 2023).

4.8 Hybrid methods for cooling

A potent two-pronged attack against heat, hybrid cooling combines techniques including liquid and air cooling. It uses air cooling for less intensive areas and tailored liquid cooling for hot regions. Data centres and other high-demand applications benefit significantly from this combination since it keeps things cooler and uses less Energy. The panel isn't seen in detail, but it's next to a multimeter on a surface. One of the most important ways to determine how efficient a solar panel is is to use a multimeter to measure electrical parameters such as voltage. To learn how single-cell solar laminates collect light and transform it into heat or Energy, researchers tested a variety of their features. According to research, the rear glass layer of glass-glass solar panels absorbs the most heat, while the overall efficiency of the panels is 70–76% (Elbreki et al. 2021).

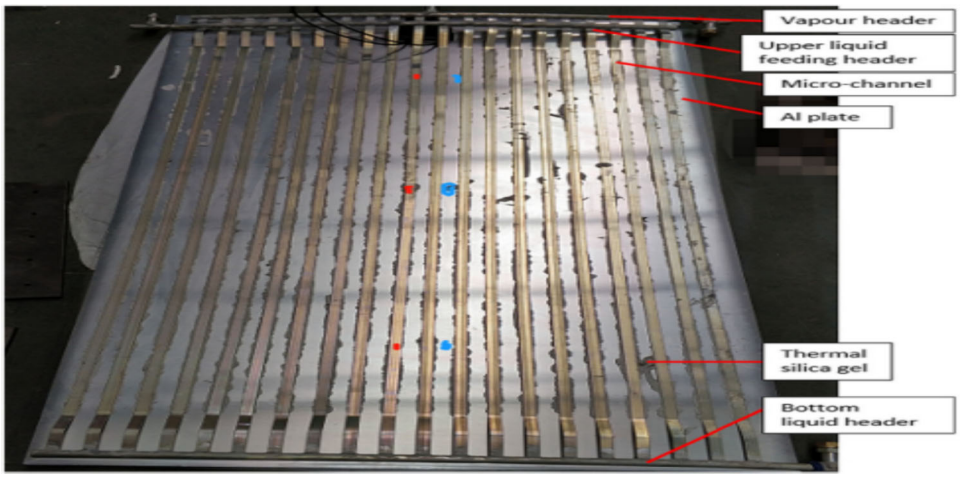


Fig. 17 Connecting solar PV panels to heat pipes (Verma et al. 2020)

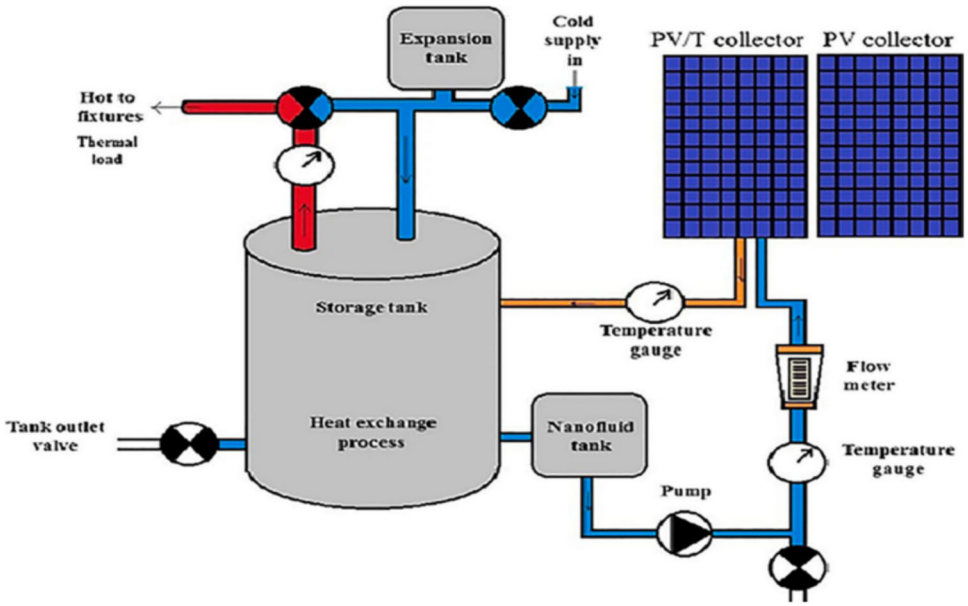


Fig. 18 Nanofluidic PV/T system schematic (Al-Waeli et al. 2017)

One method for producing cold air for HVAC systems is solar cooling, which harnesses solar Energy. Under different lighting conditions, this research contrasted cooled and uncooled solar panels. To avoid overheating-related damage, it was determined that cooling is essential. Factors such as climate, budget, and intended use of waste heat determine the optimal cooling method (Gao et al. 2023). A thermal hybrid system that uses a circulating pump was examined using a detailed mathematical model. Both the amount of sunlight collected by the system and variations in water temperature were accounted for by this model (Yang et al. 2024). While the sun's rays gradually diminish throughout the day, the power output of solar panels also decreases. To increase the efficiency of more recent solar cells, cooling them has been the primary focus of research (Chea et al. 2023). To cool solar panels, researchers tried four different approaches using water. Evaluation of efficacy was based on comparisons with a control group that did not get cooling (Verma et al. 2020). Due to the PV paradox, solar panels become less efficient as temperatures rise. Buildings can repurpose thermostats that use solar panels for cooling. The two most common forms of cooling are passive and active. When there is more sunshine, solar panels may produce more electricity (El Alami et al. 2024). The purpose of this research is to determine the best ways to cool solar panels so that they produce more electricity and use less Energy overall (Alami et al. 2025c). Uncontrollable environmental factors, such as dust and temperature, affect solar panels. A basic cooling approach is shown in Fig. 18; it does little to improve panel performance (Rajput and Yang 2018).

Solar panel efficiency is affected by several factors, both external and internal to the panel itself (Lamers et al. 2018).



Fig. 19 Methods for conducting experiments using various measuring instruments (Mutasher et al. 2020)

The efficiency of solar panels is affected by a myriad of internal and external factors. It doesn't account for factors such as room temperature, wind direction, incoming radiation rate, and dust accumulation on PV cells (Olawole et al. 2019). However, energy storage devices are the way to go when it comes to generating electric power, since solar Energy is mainly intermittent. By reflecting sunlight and sending it into space, passive midday radiative cooling (PDRC) naturally lowers surface temperatures. Present PDRC designs are inefficient, albeit they show promise (Mandal et al. 2018). Using solar thermal and PV cell technology, PV/thermal (PV/T) collectors, as depicted in Fig. 19, generate thermal and electrical Energy simultaneously (Mutasher et al. 2020).

To find the best fin design for passively cooling solar panels, scientists employed a specialised method. The study determined that heat sinks with longitudinal or lap fins function best at a fin height of 200 mm. Fifteen longitudinal fins and eighteen lap fins were the optimal configurations (Shehram et al. 2026).

Solar panels can be cooled in a variety of ways, some of which have been invented by scientists. These include utilising specific waxes, blowing air or water on them, and combining them with heat collectors. Overexposure to sunlight causes solar panels to overheat and eventually fail. Significant cooling occurs in sunny regions. This study examines 13 distinct cooling systems to determine the most effective and economical. It thinks that systems that absorb just one influence could be a wise choice (Jiao and Li 2023). Despite the potential, there is a lack of comprehensive assessments of the performance of solar PV-thermal systems in buildings (Rukman et al. 2020). To improve design and forecast the performance of future solar technologies, Fig. 20 examines operational solar power facilities (Aslam et al. 2022).

Improving PVT collector-based solar cooling systems was the focus of research. Results showed that specific PVT collector designs could increase efficiency without reducing solar panel power output (Almitani and Elamin 2023). Cool weather is optimal for solar panels. This study examined how a tiny heat pipe design—specifically, the channel width—affects its heat removal capability from solar panels. Heat dissipation was enhanced by using wider passages (Micheli et al. 2013). Potential uses for PVT collectors include building-integrated PV (BIPV). Nanofluids, phase change materials (PCM), and heat pipes have all made strides, but these technologies are still in the early stages of research and can be pretty pricey (Rukman et al. 2020). How solar PV/T systems work and how to gauge their efficiency were discussed in the introduction. Something as shown in Fig. 21 can harm these systems, leading them to produce less power (Grubišić-Čabo et al. 2018).

4.9 Nanofluid photovoltaic coolant

Suspended nanoparticles in a base fluid are used to improve heat transmission in nanofluid cooling. Nanofluids are ideal for high-performance applications that require effective cooling, as nanoparticles significantly enhance heat dissipation. Nanofluids improve efficiency, prolong component life, and optimise operating temperatures. The heat-absorption capacity of the testing fluid is enhanced by incorporating nanoparticles and fins into the pipe. Each layer's predicted heat generation is added to the solid-layer source terms in the conduction equations. The electrical performance of the cell is enhanced by cooling it with fins (Almitani and Elamin 2023). The versatility and efficiency of solar PV/T technologies are driving their rising popularity. Under certain conditions, a study showed that a flat-plate PV/T collector can accomplish radiative cooling without modifying its structural components. Two cooling methods were proposed to maximise the efficiency of a photovoltaic (PV)- powered RO desalination plant by simultaneously heating the RO feed water and cooling the PV. The outcomes of testing these cooling methods at various HTF flow rates are shown in Fig. 22 (Shalaby et al. 2022).

This research focuses on a unique solar panel sensor featuring an additional layer. Increasing the thermal and electrical outputs is the target. To effectively regulate heat, the sensor employs a novel heat exchanger that uses three coolants (Abidi et al. 2013). This under-researched review (Chunxiao et al. 2020) examines methods for reducing the temperature of solar panels to increase their efficiency.

4.10 Radiative cooling materials

An alternative to active cooling systems that rely on electricity or mechanical components is passive radiative cooling. The materials used to disperse heat into space are highly efficient infrared emitters. This technique allows heat to escape effectively at wavelengths that are transparent to the atmosphere. Radiative cooling is an exciting new paradigm in passive cooling and thermal management that could revolutionise several sectors. To conquer obstacles and reach its full potential, additional research and innovation are required.

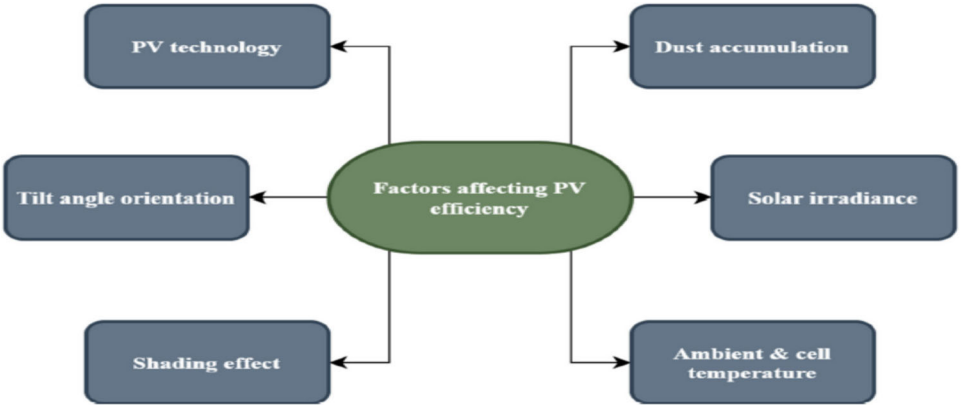


Fig. 20 Variables impacting the effectiveness of PV systems (Aslam et al. 2022)

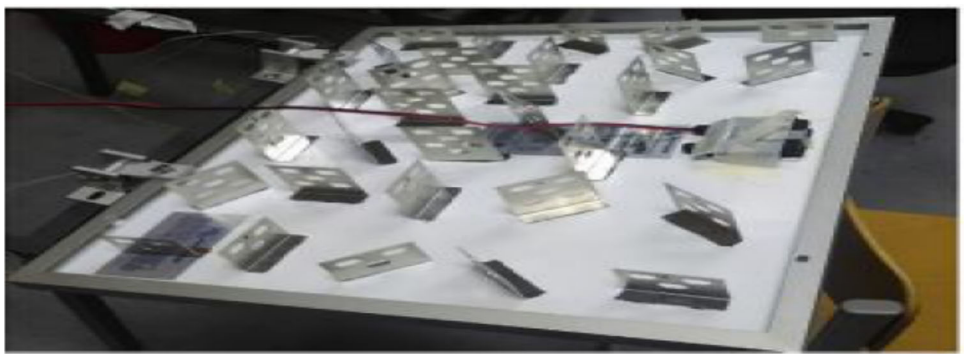


Fig. 21 Solar cell reverse side (Grubišić-Čabo et al. 2018)

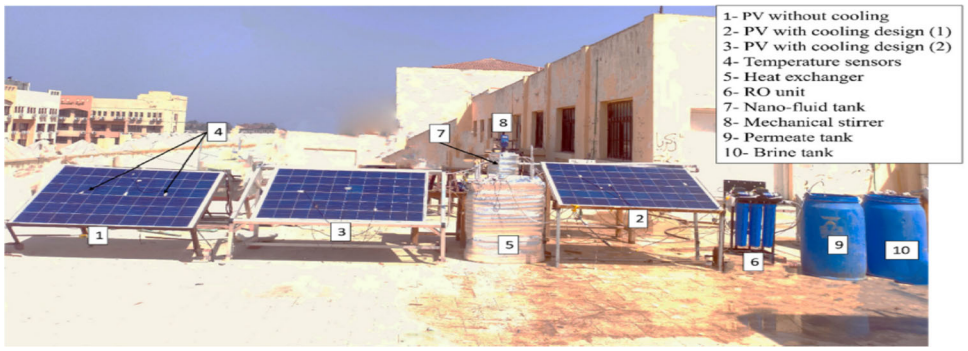


Fig. 22 Systems that combine preheating with cooling through the use of PV-RO architecture (Shalaby et al. 2022)

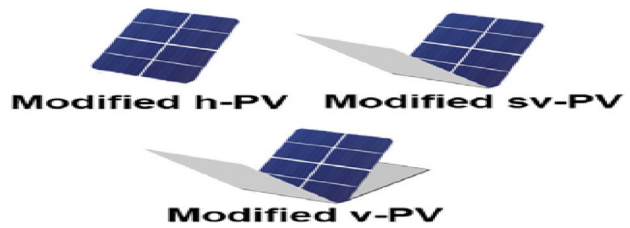


Fig. 23 The three photovoltaic systems were improved (Fang et al. 2024)

Reducing the temperature of solar panels using radiant cooling is an energy-efficient way. Figure 23 shows the limitations of present materials in PV modules due to their thermal characteristics (Fang et al. 2024).

Improved radiative cooling of solar cells is achieved by micro-grating silica photonic coolers. As a result, the thermal emissivity increases to more than 0.9, while solar reflectance decreases (Zhao et al. 2022). The key to achieving peak power is keeping solar panels cool. Panels that get too hot lose some of their effectiveness, but with the help of fans or other advanced cooling systems, solar systems can produce even more Energy. For electronics and industrial processes to run efficiently, lowering temperatures is crucial. Better performance, lower energy consumption, and longer system life are all benefits of using cooling solutions to prevent overheating, such as heat sinks or sophisticated fluids. This not only helps save money but also helps the Environment. Solar panels remain content with efficient cooling. When properly cooled (via fans or ventilation), hot panels survive longer, require

Table 1 Comprehensive Comparative Analysis of PV Cooling Methods

Parameter	Passive cooling	Active cooling	Hybrid cooling
Performance improvement	Moderate (5–12% increase depending on design)	High (10–25% depending on air/water flow rate)	High (12–30% due to the combined effect)
Temperature reduction	5–15 °C	10–25 °C	12–28 °C
Capital cost	Low	High	Medium–High
Operational cost	Zero or negligible	High (Energy for pumps/fans, water consumption)	Medium (reduced power usage compared to active)
Maintenance requirements	Very low	Frequent (cleaning, pump/fan service)	Moderate
Energy consumption	None	Continuous; reduces net gain	Low–moderate, depending on design
Water requirement	None	High (for water spray/evaporative cooling)	Medium (reduced frequency or closed loop)
Material requirements	Low; simple structures (fins, heat sinks, PCMs)	High (fans, pumps, piping, sensors)	Medium (PCMs + low-power cooling components)
Environmental impact	Very low; sustainable; minimal emissions	Moderate to high; higher carbon footprint	Medium; depends on material and energy mix
system complexity	Simple	Complex	Medium–High
reliability & lifetime	High reliability; long life	Dependent on mechanical parts; shorter lifespan	Moderate reliability; component-dependent
Suitability for large-scale PV farms	Highly suitable	Limited unless benefits outweigh the cost	Selectively suitable
Climate suitability	Best for hot-dry regions where passive airflow works well	Suitable for high-irradiance, extremely hot climates	Ideal in areas demanding enhanced cooling
Scalability	Easy to scale	Expensive and complex to scale	Moderate scalability

less maintenance, and generate more Energy in the long run. Solar electricity becomes more sustainable and cost-effective in this way.

5 Economy and the environment

Both the financial and environmental impacts of solar thermal cooling structures must be considered when evaluating them. Passive cooling is the most cost-effective option because it uses less Energy and costs less up front. When it comes to active cooling systems, the price of fluid circulation machinery, pumps, and fans is higher. Their increased energy use suggests that the net electrical benefit may increase slightly. In areas with limited water, using it for cooling also raises sustainability concerns. Because it uses fewer resources, passive approaches are better for the Environment. Material selection affects recycling rates or life-cycle emissions, especially for PCMs and heatsink structures. When alternatives to fossil fuels are unavailable, active systems' carbon emissions can increase. The most cost-effective choice is passive cooling. The performance gains must exceed the costs and environmental considerations to activate hybrid systems.

5.1 Effects on the environment and long-term viability

Assessing the environmental and long-term impacts of PV cooling systems is of the utmost importance. The best cooling system for the Environment is a passive one since it doesn't produce any additional energy or pollutants. On the other hand, active cooling systems could lead to higher energy consumption and carbon emissions if they don't run on renewable Energy. Other issues with water-based cooling that affect the Environment include the use of clean water, water drainage, and the potential for waste. The carbon footprint of a building is influenced by its material selection throughout its lifetime, particularly for metals, polymers, and phase-changing materials. Reduced carbon emissions and end-of-life waste are two ways in which bio-based or recyclable goods improve sustainability. Although hybrid systems provide better temperature control, they may require more Energy and materials. As a whole, sustainable heating strategies should follow the principles of the circular economy, minimise their environmental impact, and make the most efficient use of resources.

5.2 Challenges and future directions

Striking a balance among efficiency gains, costs, energy demand, and material sustainability is a challenge for PV cooling technologies. The high operational energy use, water consumption, and maintenance needs of active cooling systems limit their superior temperature control. Despite being more environmentally friendly, passive solutions don't always offer enough cooling when temperatures are really high. Problems with material deterioration, reliability over time, and integration complexity further limit widespread use. Research in the future should centre on improving thermal performance without adding to the environmental burden by developing advanced passive materials, such as high-conductivity composites and bio-based PCMs. Adaptive control and real-time sensors allow innovative cooling systems to enhance energy efficiency. In high-irradiance areas, hybrid systems that integrate passive and low-power active techniques show potential. Life-cycle assessments must guide innovation to guarantee environmental sustainability. All things considered, PV cooling solutions of the future should be long-lasting, affordable, and scalable. A thorough comparison of PV cooling methods is shown in Table 1.

6 Conclusion

While photovoltaic cooling systems have come a long way, It still have a way to go before they can solve thermal management problems. New cooling materials, advancements in intelligent control technology, and adaptive thermal techniques can improve the system's accuracy, efficiency, and scalability. If it wants to keep solar power plants' stability and performance from deteriorating due to heat, it needs to invest more in R&D. If it wishes to make additional strides in this domain, it requires more collaborative research initiatives that bridge many disciplines, as well as improved means of exchanging data. In view of the growing need for clean Energy, it is more important than ever to find affordable, environmentally friendly PV cooling solutions so the world can transition to a cleaner energy future. Passive and hybrid cooling systems should be the norm in the future, as they combine the advantages of traditional cooling with modern environmental standards at a reasonable price. Advanced heat sinks, bio-based PCMs, and better coatings could increase passive cooling. Enhanced efficiency with significantly lowered energy consumption is achievable with the combination of low-power hybrid systems, smart sensors, and adaptive control. Reusable and recyclable materials are preferable for the sake of sustainability in the long run. These advancements have made it possible to get PV cooling solutions that are both more effective and reliable.

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Declarations

Conflict of interest The authors declare no competing interests.

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